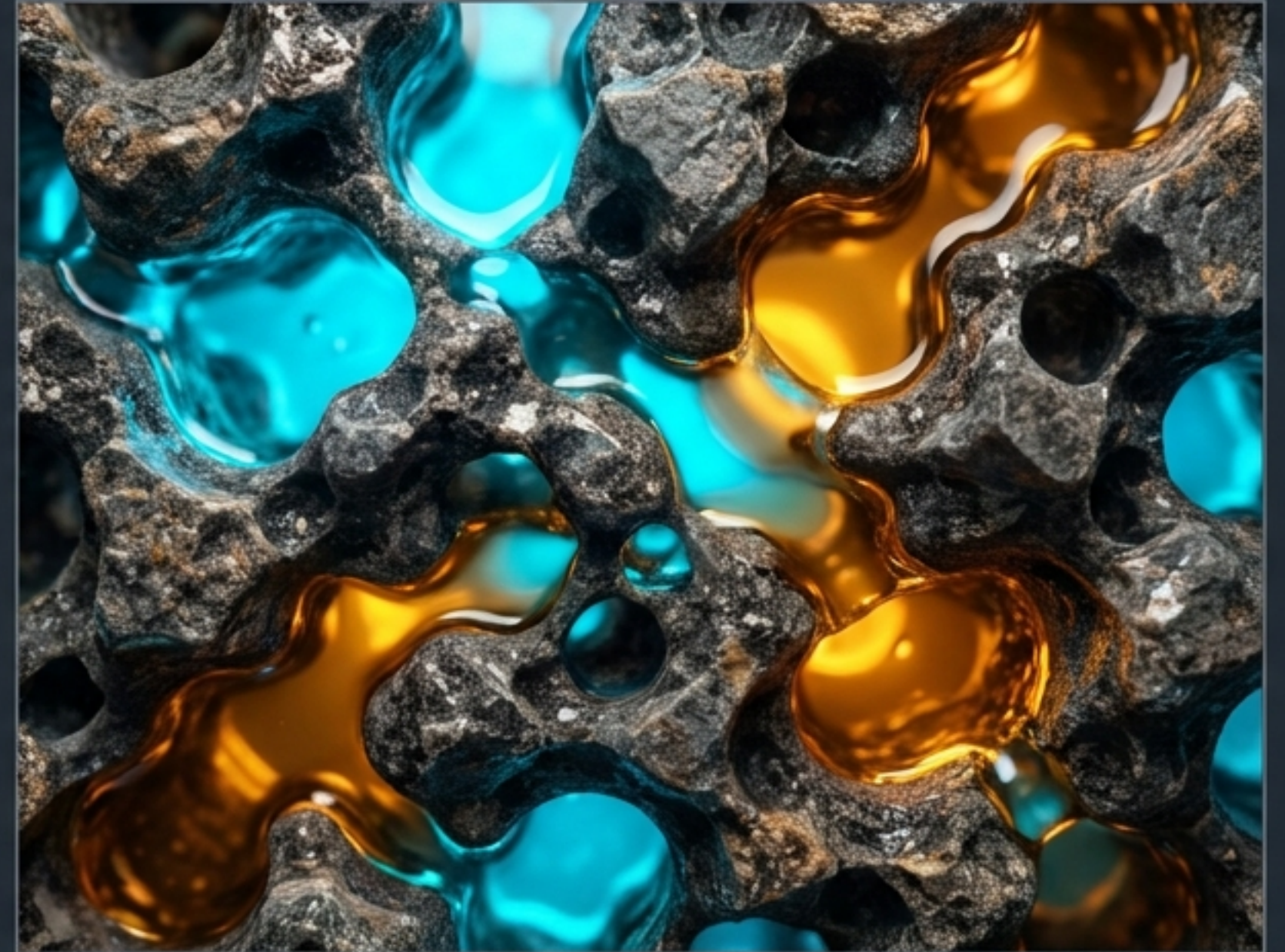
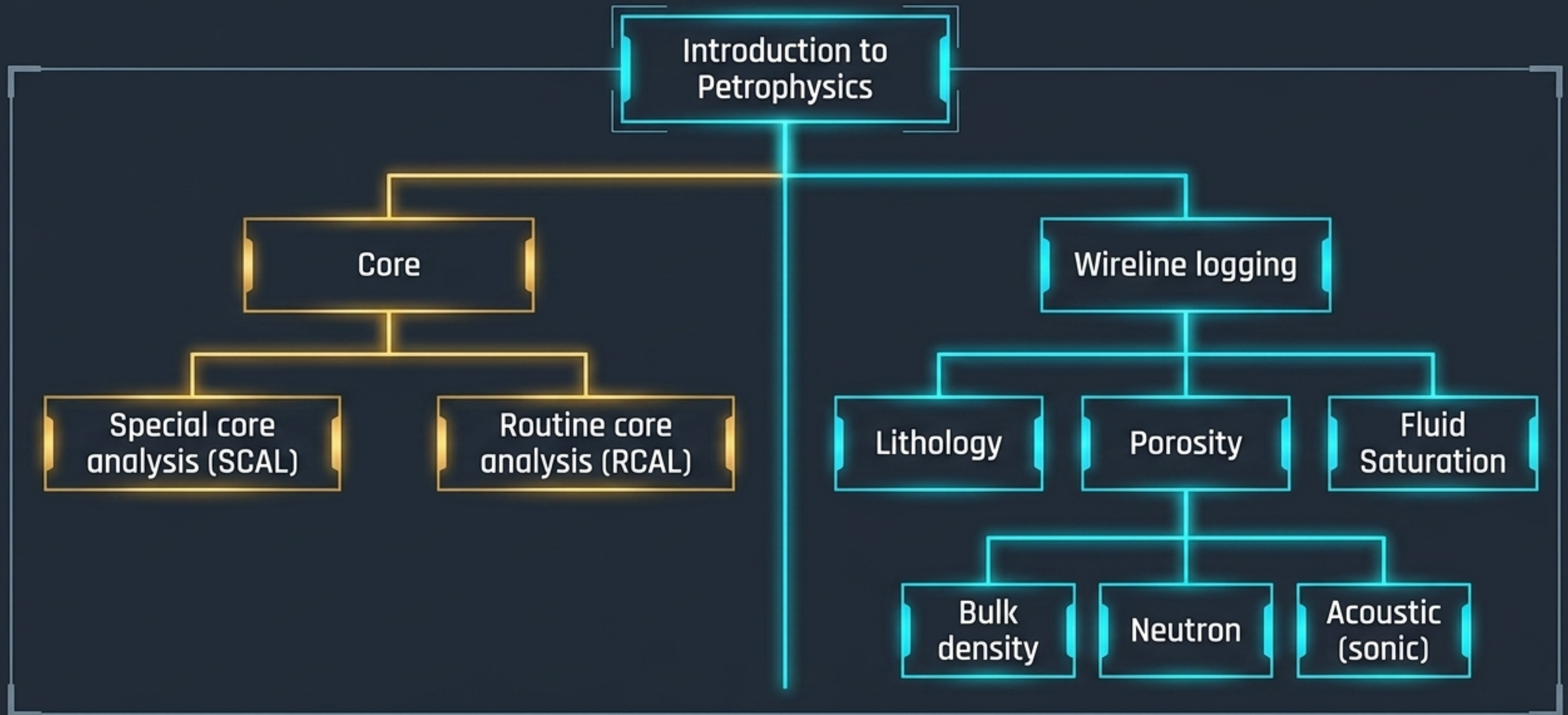


Introduction to Petrophysics: Understanding Rock and Fluid Properties



Prepared by Engineer sajjad k.hano

The Subsurface Blueprint: Dual Approaches to Reservoir Characterization



Pillar I: Core Analysis

Direct measurement of rock properties via physical extraction and laboratory testing.



Routine Core Analysis (RCAL): Establishing the Baselines



Porosity

Total void space available for fluid storage.



Grain Density

Mass-to-volume ratio of the rock matrix.



Permeability

Capacity of the rock to transmit fluids.



Fluid Saturation

Fraction of pore volume occupied by specific fluids.

Special Core Analysis (SCAL): Advanced Dynamic Properties



Electrical Properties

Formation resistivity factor and resistivity index.



Wettability

Rock surface preference for water versus oil.



Capillary Pressure

Interaction forces between rock, water, and hydrocarbons.



Relative Permeability

Multi-phase flow dynamics and fluid impedance.

Analytical Matrix: Routine vs. Special Core Analysis

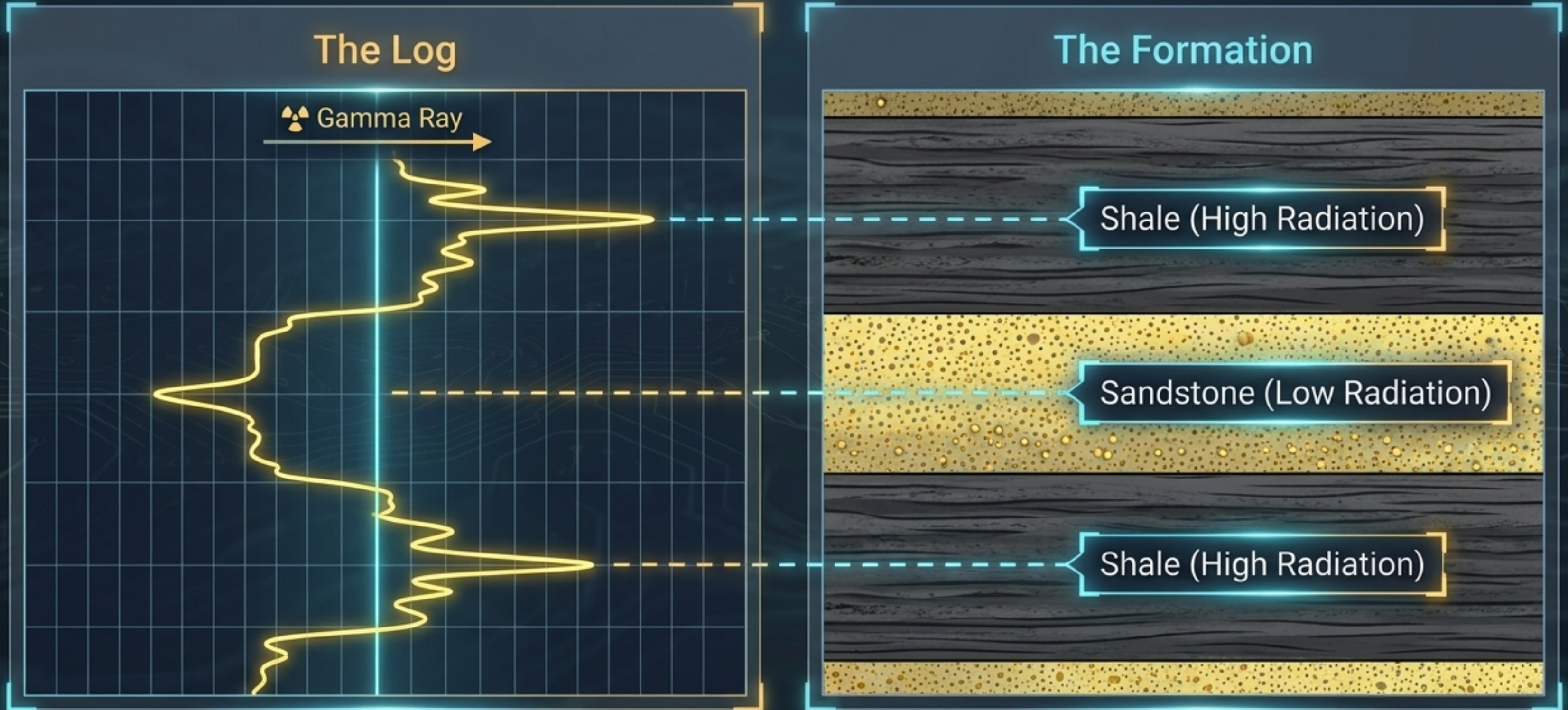
	Routine Core Analysis (RCAL)	Special Core Analysis (SCAL)
Scope	Basic, fundamental volumetric properties.	Advanced, specialized multi-phase properties.
Focus	Single-phase static measurements.	Multi-phase, dynamic flow interactions.
Turnaround Time	Days to weeks.	Months (requires complex fluid equilibrium).
Key Deliverables	Porosity, absolute permeability, grain density. 	Capillary pressure curves, relative permeability curves, wettability indices. 

Pillar II: Wireline Logging

Continuous, in-situ evaluation
of formation properties via
remote sensing.



Lithology: Identifying Rock Types via Gamma Ray



Porosity Logging: The Measurement Triad

Bulk Density



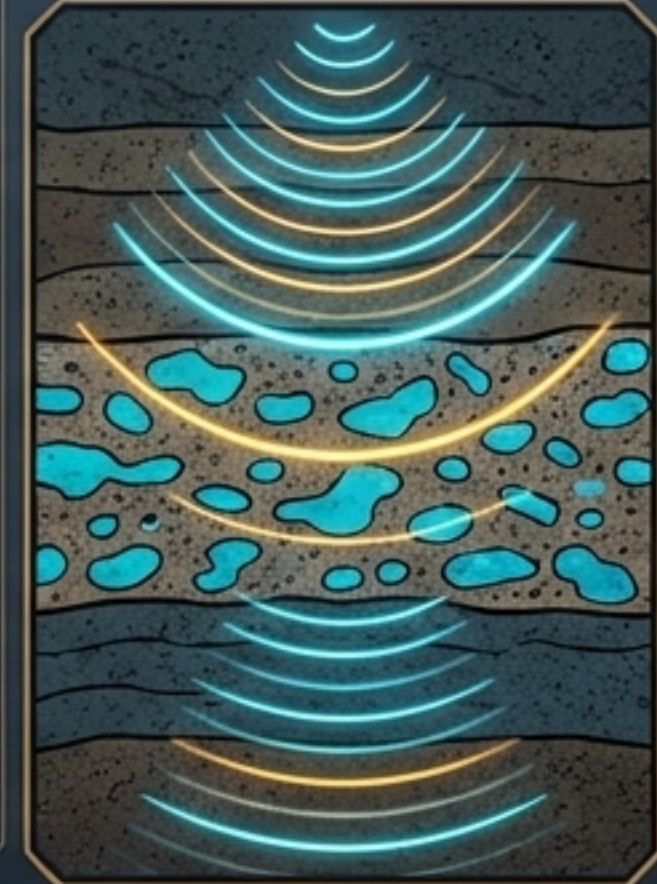
Measures overall formation density via gamma ray scattering to calculate true porosity.

Neutron



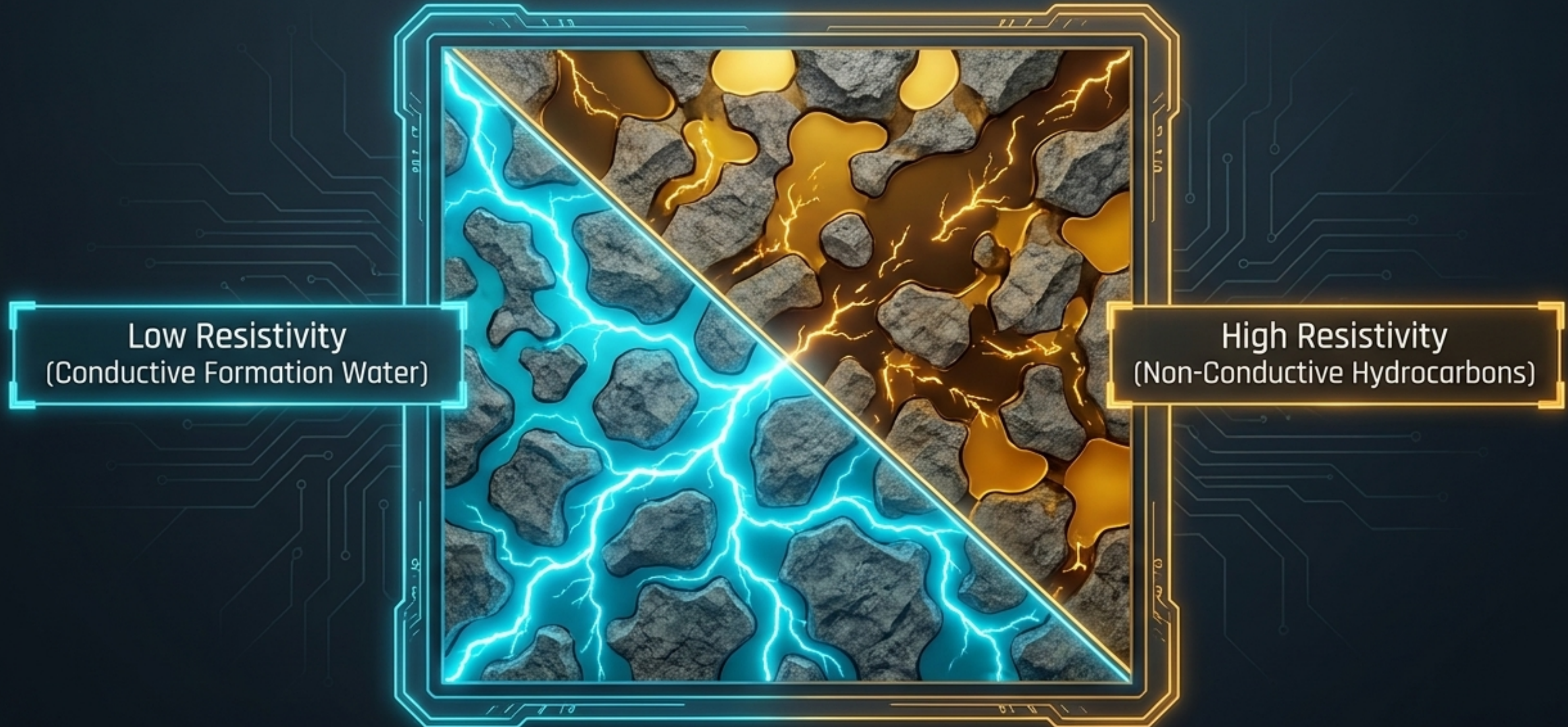
Measures hydrogen index; high hydrogen concentration indicates high fluid volume and porosity.

Acoustic (Sonic)

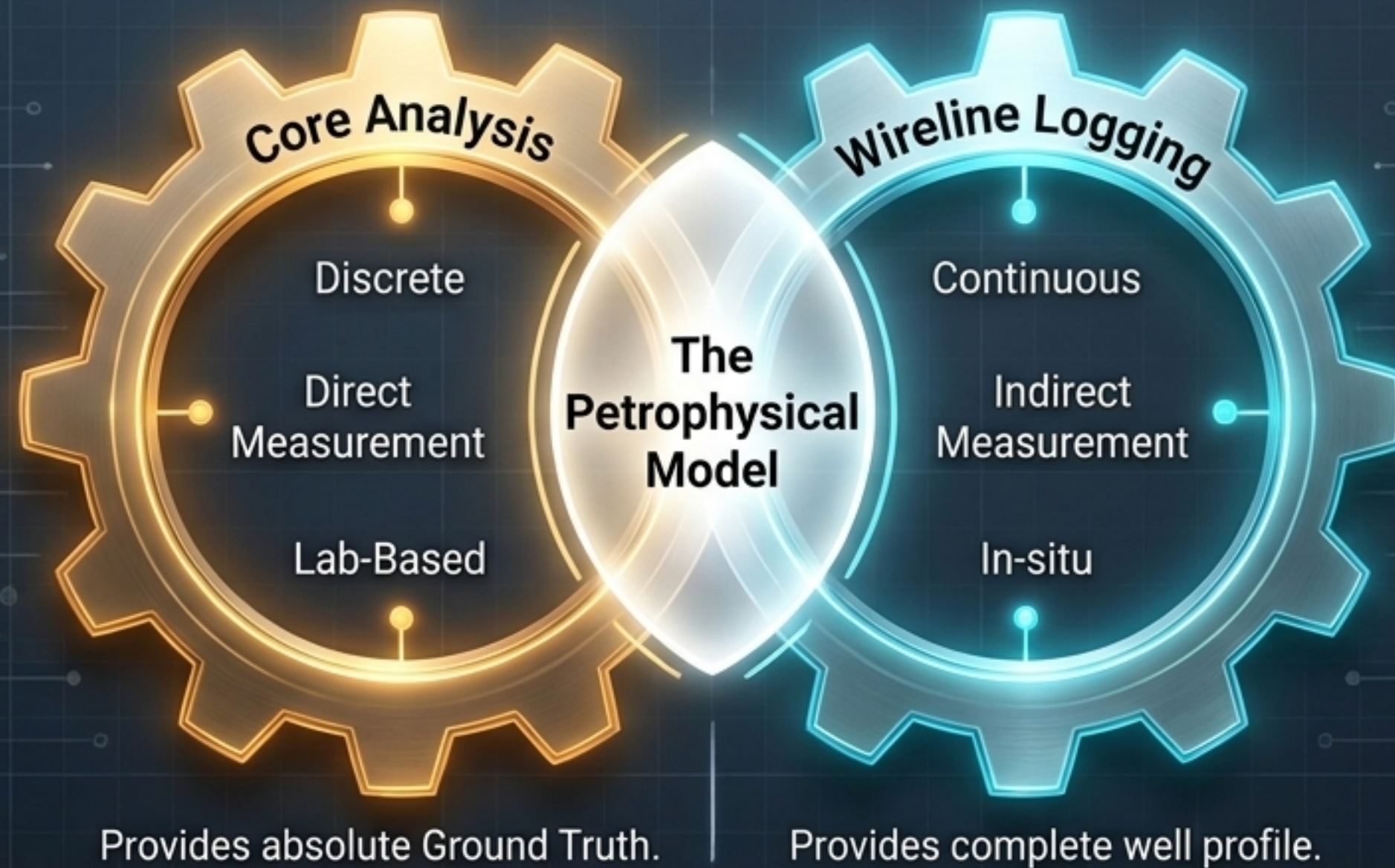


Measures compressional wave travel time; sound waves travel slower in fluid-filled porous rock than in a solid matrix.

Fluid Saturation: Mapping Hydrocarbons via Resistivity



The Petrophysical Synthesis: Calibrating the Subsurface Model



Core data validates and calibrates the mathematical algorithms used to interpret continuous wireline logs, ensuring accurate hydrocarbon volume estimations.